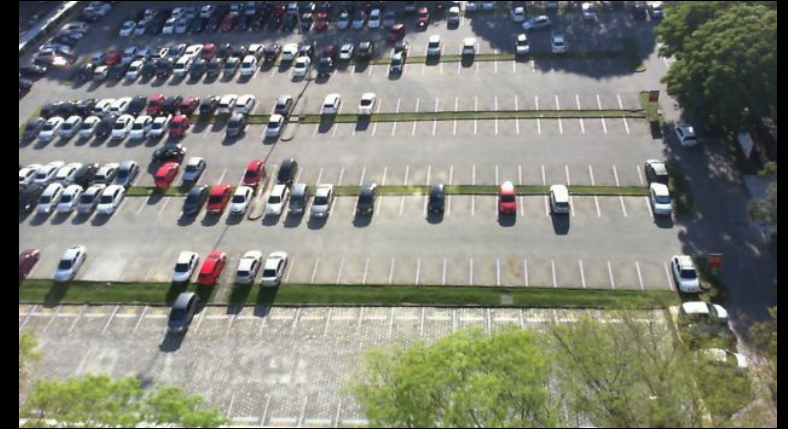


Toward Parking Spot Occupancy Recognition: A Self-Supervised Approach



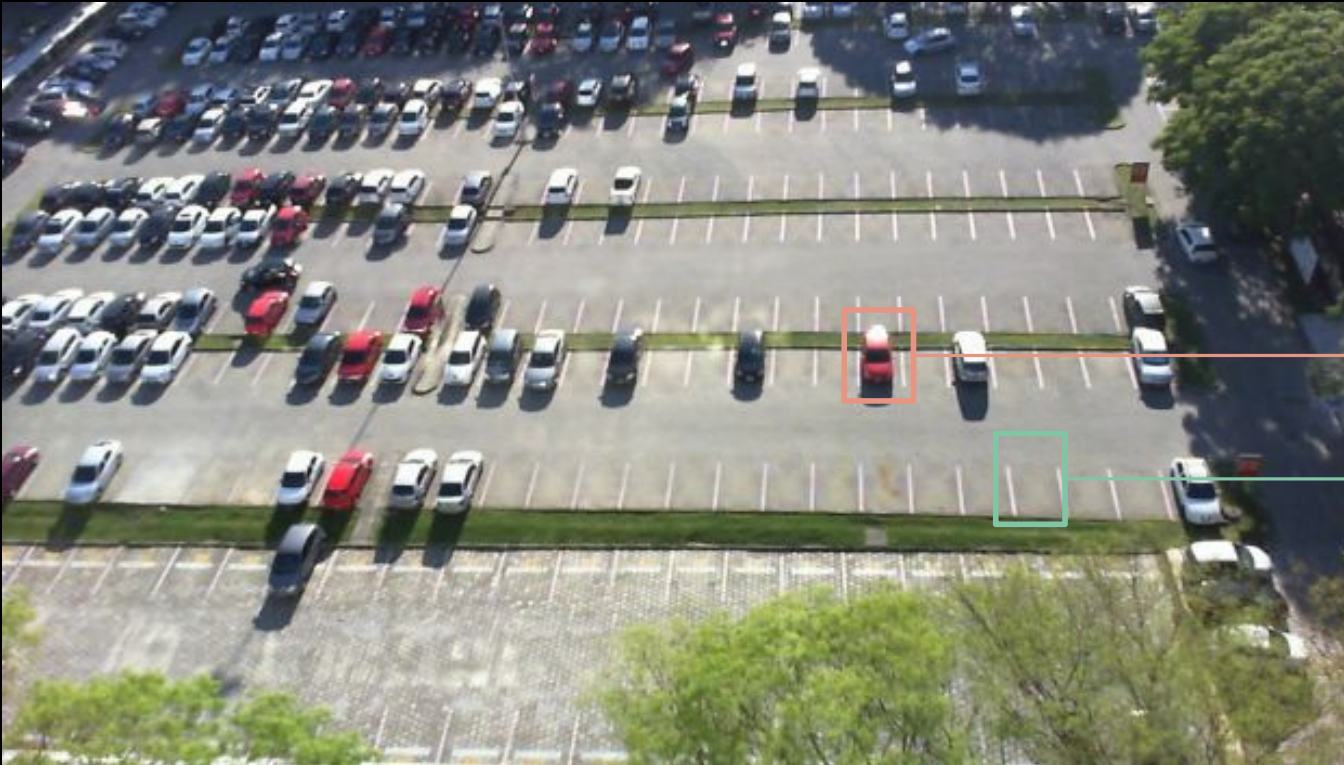
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Problem

Input: a cropped parking-space image. Output: empty or occupied.



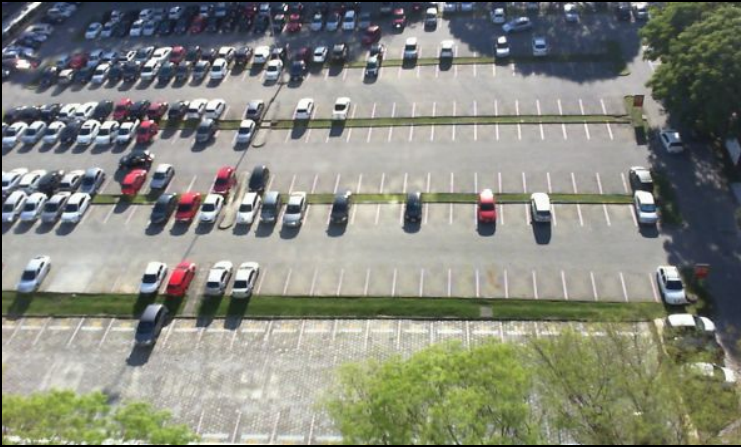
Occupied



Empty

The hard part is the next parking lot

Variation in viewpoint, lighting, and weather.



PKLot



CNRPark-EXT



PLds

Labeled source environments

New target environment

No occupancy labels

Relabeling each new deployment is the bottleneck.

Literature: high in-site accuracy, bad transfer

STUDY	WITH TARGET LABELS	WITHOUT TARGET LABELS
[1] de Almeida et al. (2015)	99%	89%
[2] Amato et al. (2017)	97.71%	83.83%
[3] Gregor et al. (2019)	98.42%	79.21%
[4] Wang et al. (2023)	99.62%	91.21%

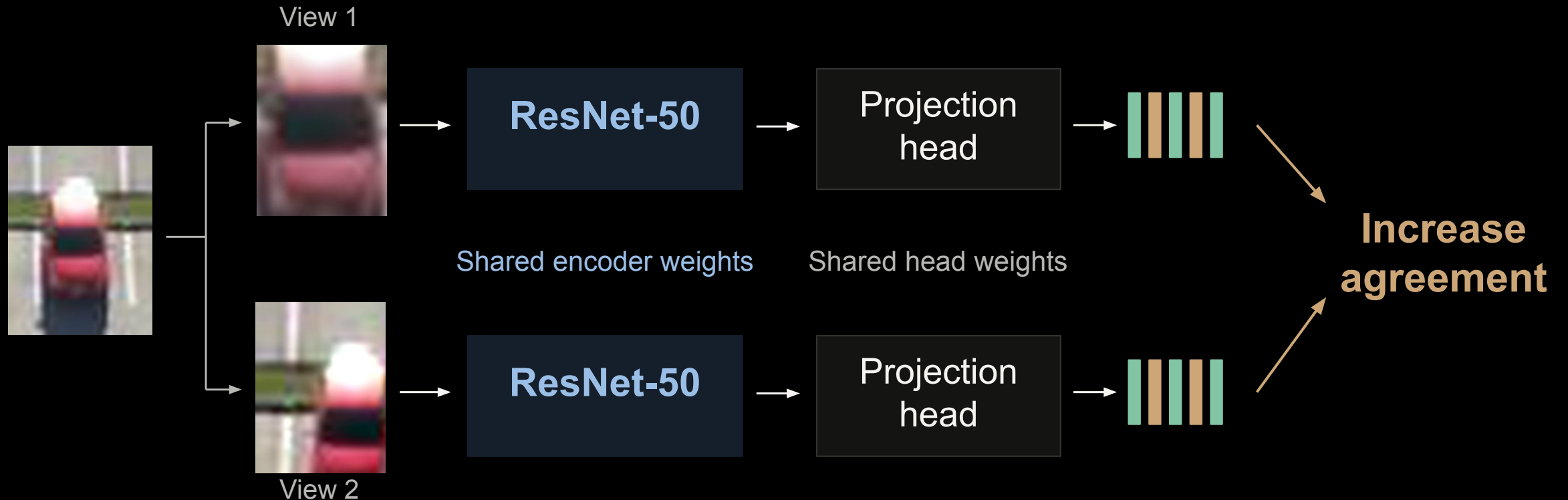
Solution

Specialize models with unlabeled data from the target camera.

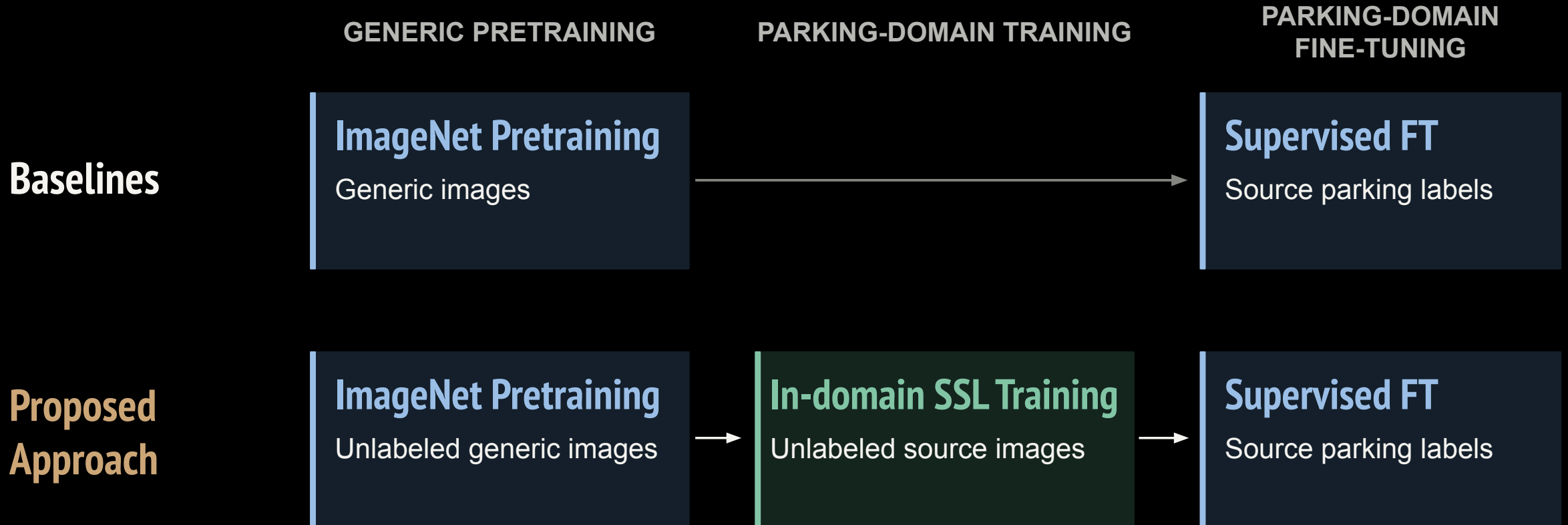
Deploy generic classifiers → collect unlabeled target data → specialize

How to specialize with unlabeled data?

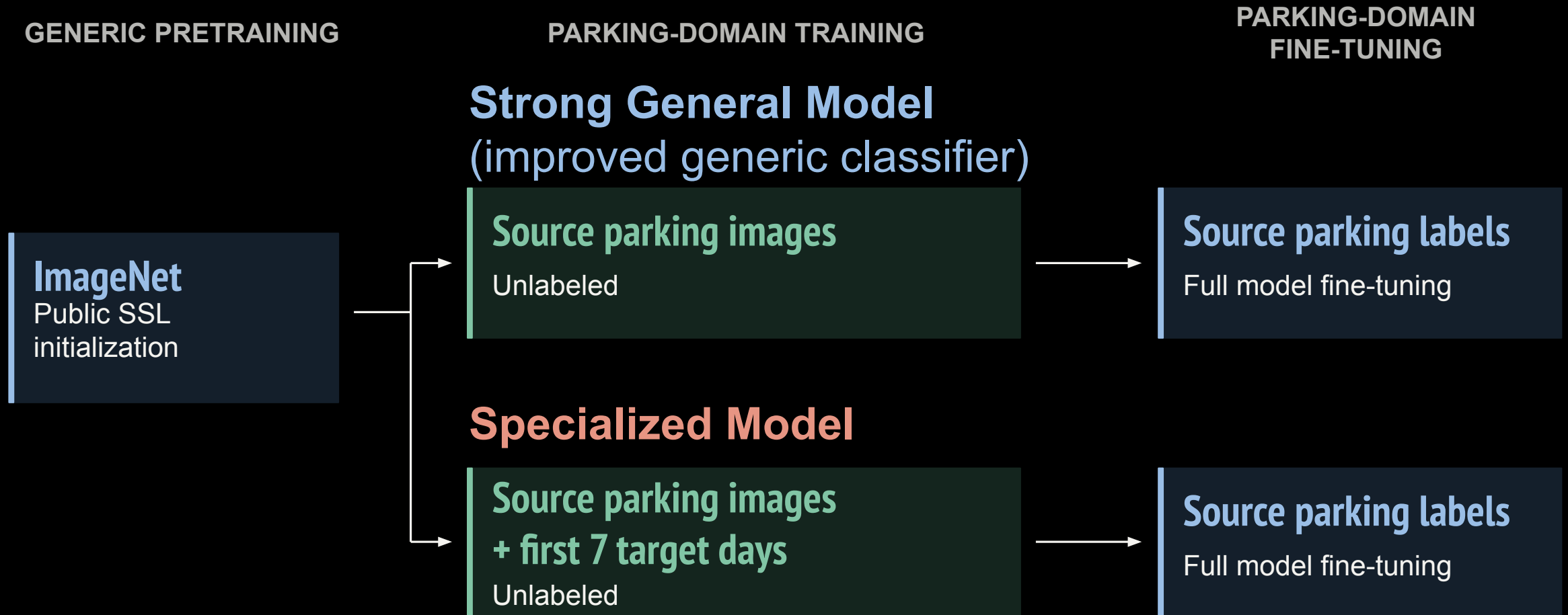
SimCLR



Method



Two model configurations



Deploy immediately. Specialize after seven days.

DAYS 1–7

Strong General Model

improved generic classifier.
Collect target images.



AFTER COLLECTION

Train the Specialized Model

Mix source images with
unlabeled target images.



REMAINING DAYS

Specialized Model

Classify occupancy
in the remaining days.

Train on two datasets. Test on the third.



PKLot

1,199,857 annotations



CNRPark-EXT

165,513 annotations



PLds

104,728 annotations

PKLot + CNRPark-EXT

→ **PLds**

PKLot + PLds

→ **CNRPark-EXT**

CNRPark-EXT + PLds

→ **PKLot**

Four configurations

Configuration	ImageNet pretraining	Parking-domain SSL	Final supervised fine-tuning
Supervised Baseline	Supervised	-	Source labels
Self-supervised Baseline	SimCLR	-	Source labels
Strong General Model	SimCLR	SimCLR on source images	Source labels
Specialized Model	SimCLR	SimCLR on source images + first 7 target days	Source labels

Results

Global averages reported in our evaluation protocol

Self-supervised Baseline

Supervised Baseline

Strong General Model

**Strong General Model &
Specialized Model**

97.0%

97.1%

97.2%

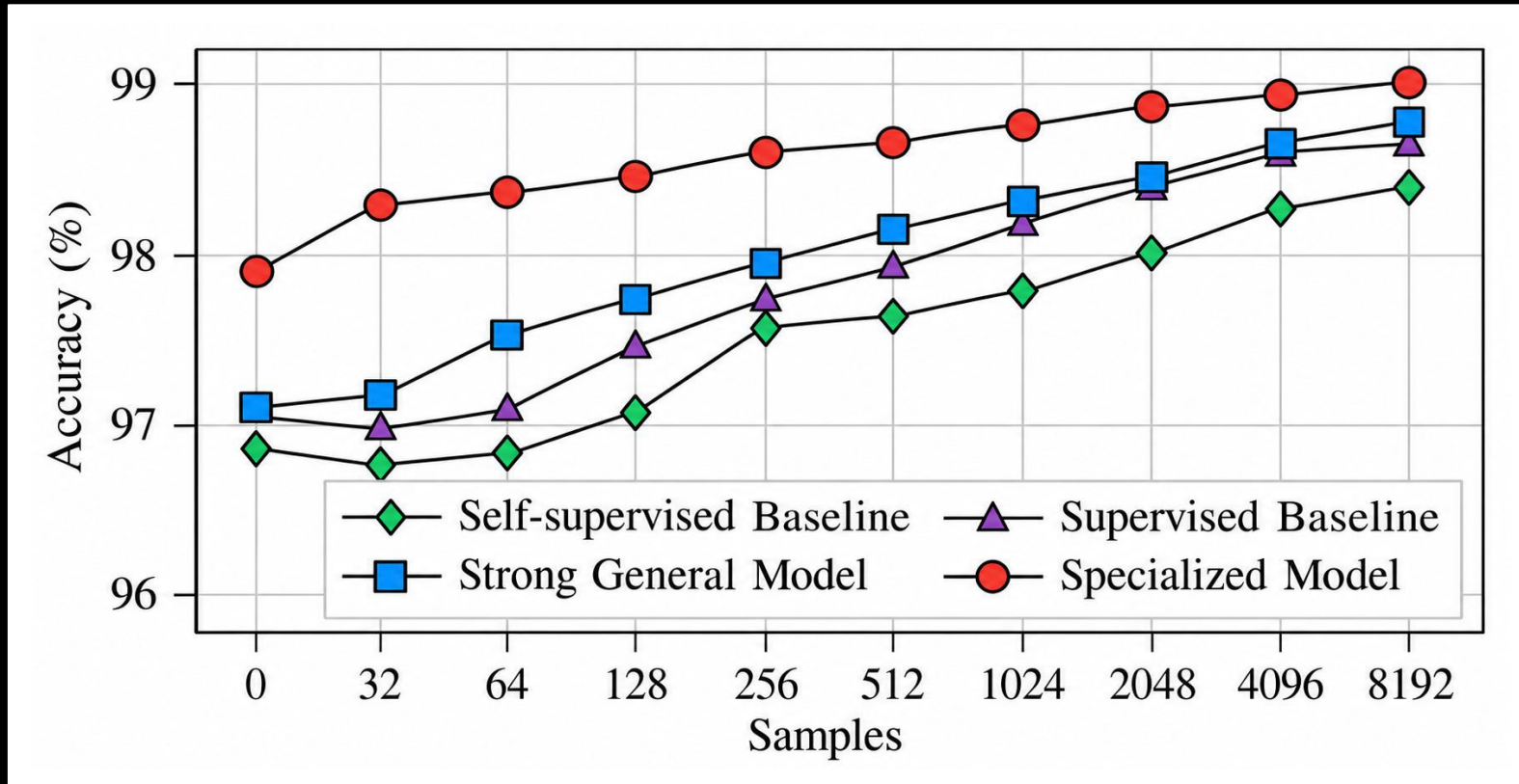
97.8%

97.0 97.2 97.4 97.6 97.8 98.0

Accuracy

97.8% with no target occupancy labels

The specialized model also uses labels efficiently



Optional target labels

Labels are sampled from the first 7 days.

Cost

~25 h

Training per model

Single GPU
Strong General or Specialized

0.2282 s

**Inference time per
parking spot**

Raspberry Pi 5

*Strong General Model is trained once
and can perform in every parking lot

Adapt to a new lot without new occupancy labels.



Paper



Code & Models



DSBD

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